

Mesh: A Token-Efficient, Graph-Native Logic for Repository-Scale Code Agents

Edgar Elmo Baumann · Philipp Nils Horn

Mesh · try-mesh.com

edgar.baumann@try-mesh.com · philipp.horn@try-mesh.com

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SUMMARY

Mesh is a pre-inference control plane for repository-scale code agents. The observation behind it: most of a coding agent's token budget goes to *finding* the right code, not reasoning about it. Mesh materializes the repository locally into a field-weighted lexical index, dense code-embedding vectors, and a deterministic repository intelligence graph (RIG), then assembles *budgeted, cited evidence before every model turn* — rather than letting context accumulate as an unbounded side effect of the agent loop.

Headline results. On a grounded suite of **229 localization tasks across six external repositories** (gold derived mechanically from the code), intent-routed Mesh reaches **Hit@1 79.9% / Hit@8 96.9%**. In a controlled head-to-head on identical pool, queries, and gold against the strongest retrieval baselines — full-file BM25, NVIDIA nv-embedcode-7B, and AWS Bedrock Cohere Embed v4 + Rerank 3.5 — Mesh wins **79.9% vs 32.8%** for the best single-method baseline (2.4×, McNemar $p < 0.001$; Cohen's $h = 0.99$). Against a competent grep-based agent it localizes at **−69.7% tokens per query** and **−55.5% over multi-turn sessions**, and a same-model scaffold study (Haiku 4.5 and Opus 4.8 in both arms) shows **−20% to −26% fewer output tokens and tool calls at equal recall**.

Keywords: hybrid retrieval · repository intelligence graph · context budgeting · cost-normalized agents
· pre-inference control plane

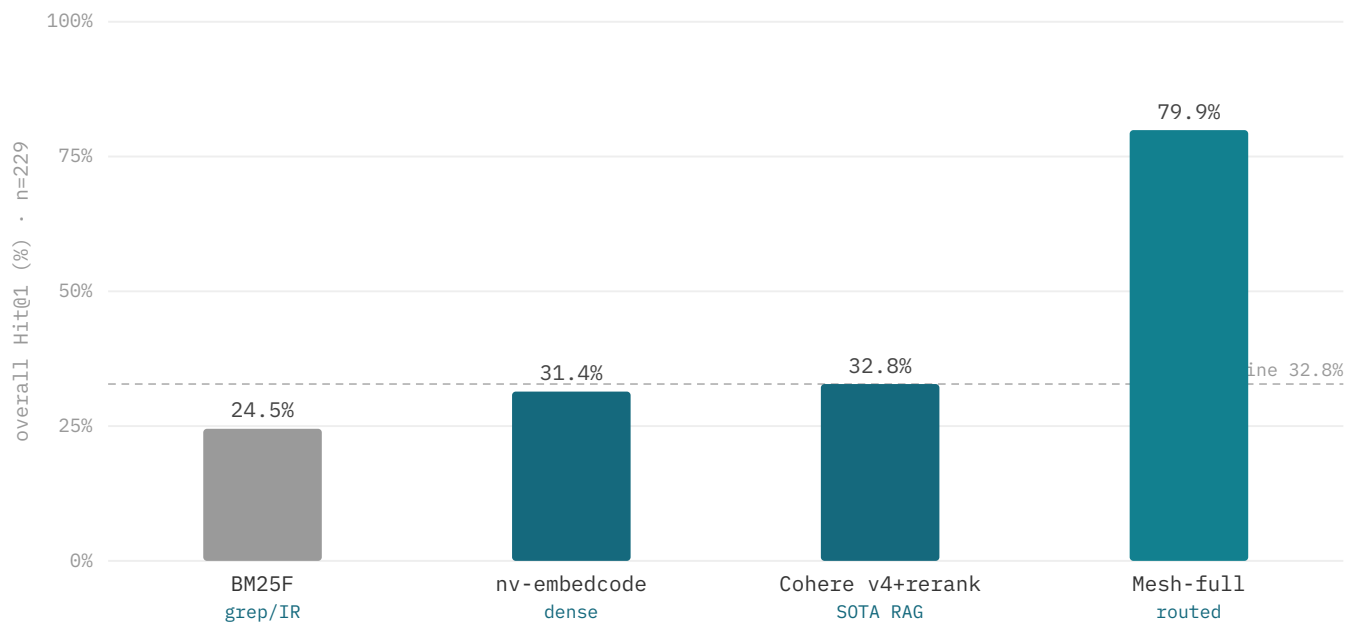


Figure 1. Overall Hit@1 on the n=229 suite: intent-routed Mesh 79.9% vs the best single-method baseline 32.8% (Cohere Embed v4 + Rerank 3.5), on identical pool, queries, and gold.

The full paper develops the architecture as five independently-ablatable layers (representation → evidence → context → economics → runtime), a formal constrained-utility model of the agent loop, the complete evaluation contract (retrieval ablation, needle-in-a-haystack, multi-turn economics, the SOTA baseline head-to-head, and a same-model scaffold study), and an honest treatment of threats to validity. We would welcome your feedback on the framing and the evaluation design.

— Edgar Elmo Baumann & Philipp Nils Horn · Mesh · try-mesh.com · edgar.baumann@try-mesh.com